# Torque Crank Angle Curve of a SI Engine

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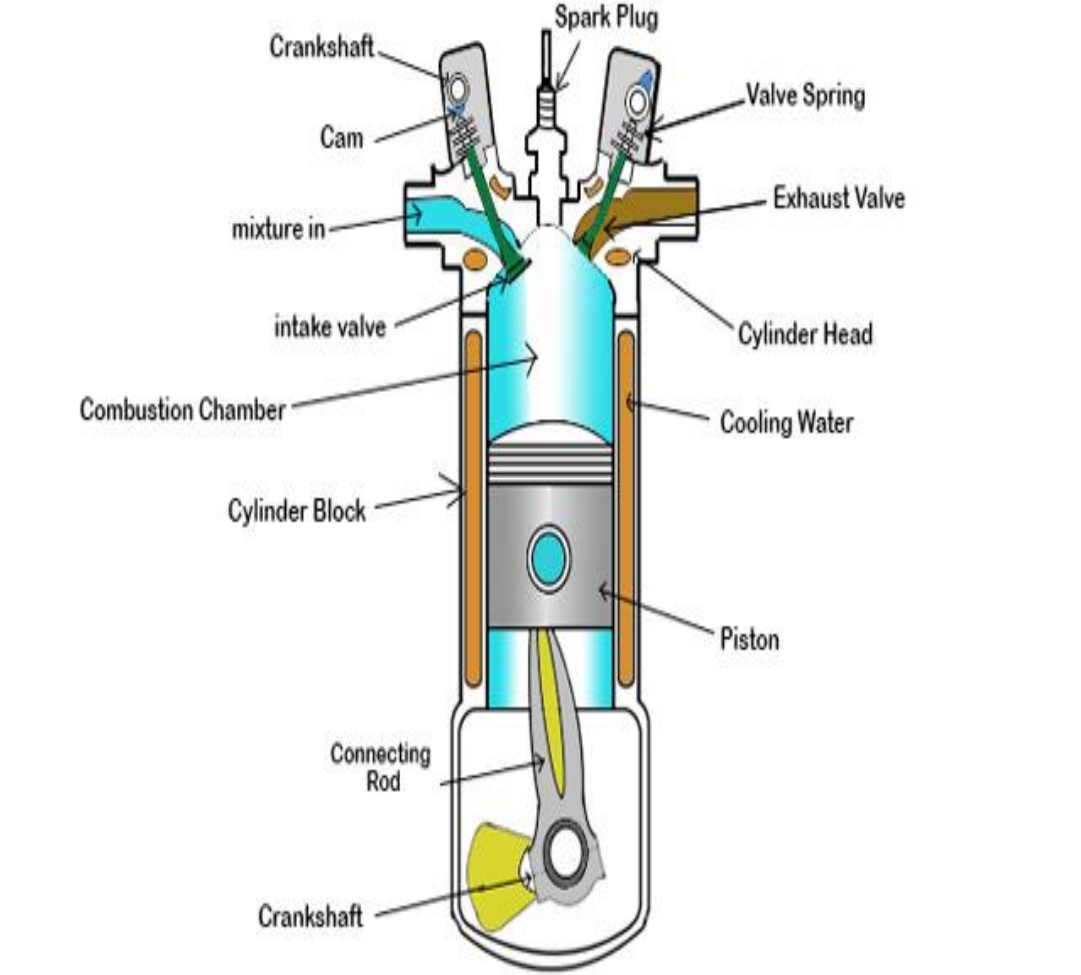
# WHAT WE WILL COVER :

1. Introduction
2. Procedure in Lab
3. Explain Turning Moment Diagram
4. Flywheel (Crank Effort Diagram)
5. Cycle Analysis – What Is happening exactly ?
6. Simulation Video
7. Simulation Analysis

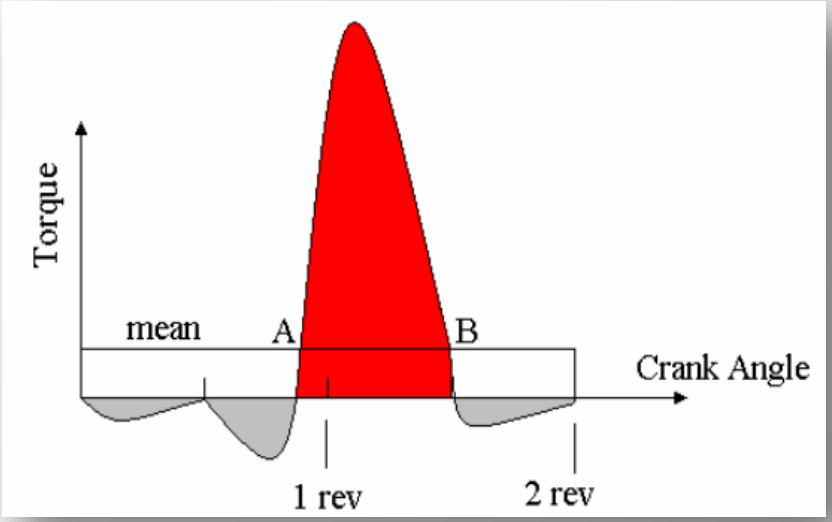


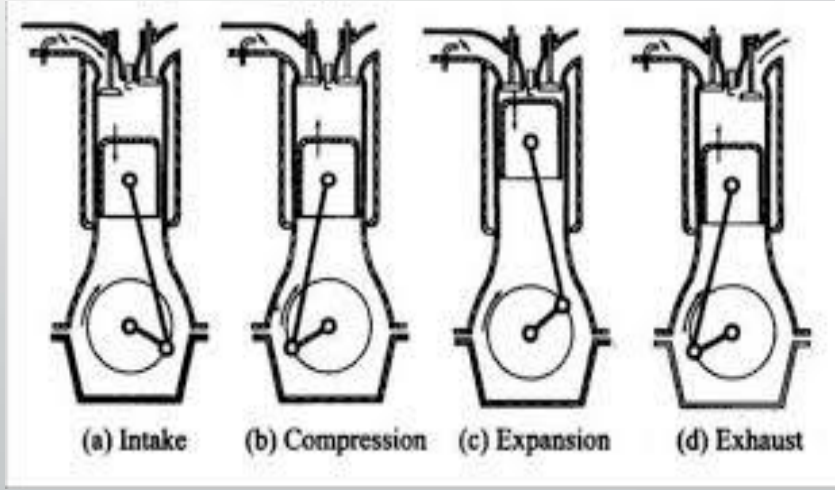

SI Engine Parts

# INTRODUCTION

The complete cycle takes two revolutions divided into four equal strokes. It is known that in a four stroke cycle SI engine. Since the pressure inside the engine cylinder is less than the atmospheric pressure during suction stroke, therefore negative loop (0-π crank angle) is formed (as shown in crank angle 0-π). During compression stroke, the work is done on the gases therefore, negative loop is obtained. During expansion or cracking stroke, the fuel burns and gases (air-fuel mixture) expand therefore large positive loop is formed. During exhaust stroke, the work is done on the gases, therefore negative loop is formed (3π -4π) crank angle.

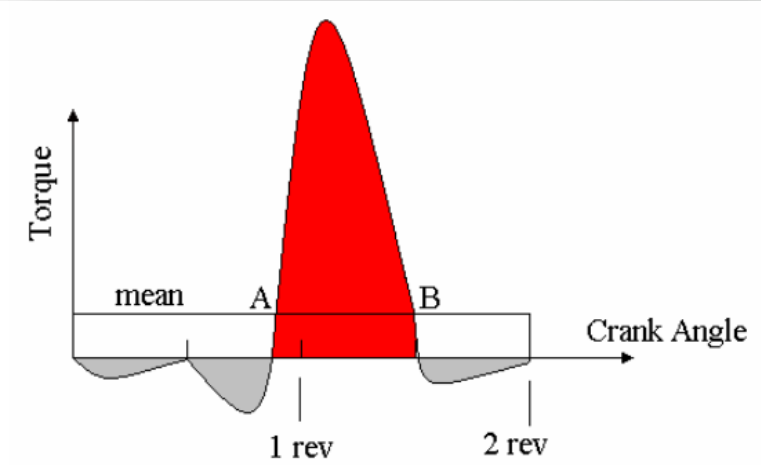
**Consider a graph of torque against crank angle for a typical 4 stroke engine:**

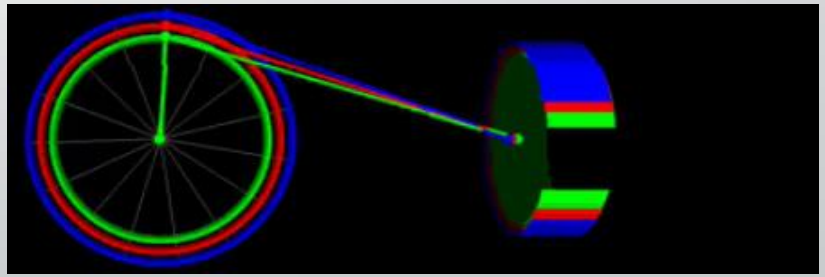


# INTRODUCTION

The complete cycle takes two revolutions divided into four equal strokes. One revolution is 2π radians so it takes 4 π radians to complete the cycle. Only the red part of the diagram represents energy given out (the power stroke). The induction, compression and exhaust strokes require negative torque as energy is required to do these. If the machine is to be a successful engine, the area of the red part must be larger than the areas of the grey parts giving a positive mean torque. The area above and below the mean line must be equal by definition of mean. The area of the rectangle formed between the mean and zero must be the work output of the engine per cycle.



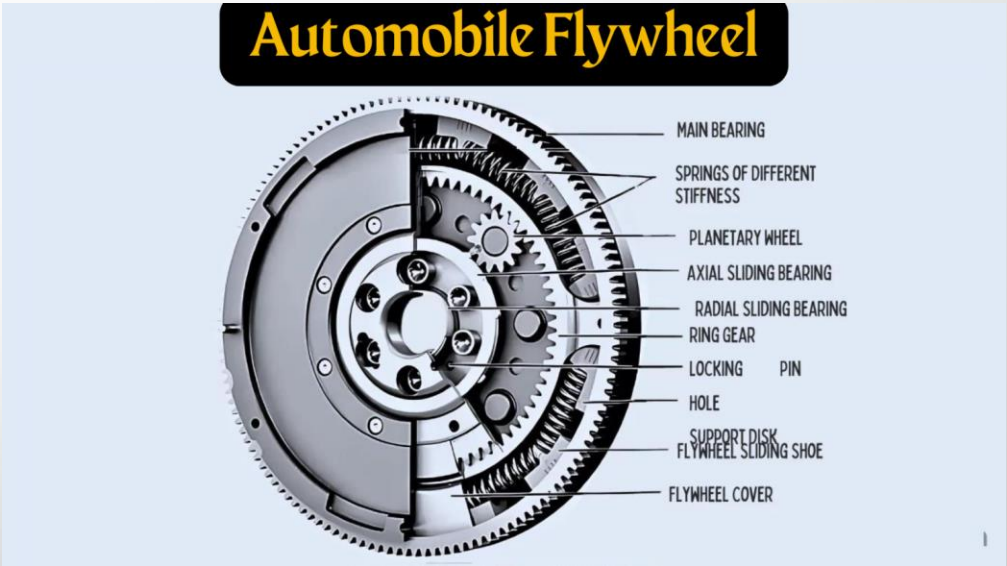



# PROCEDURE IN LAB

1. For working from the lab , one has to switch on the water pump, the motor, fuel supply, and the control panel.
2. Once the computer starts, you need to open the already installed EPA software. Then you have to crank the engine.
3. Once the engine starts go to Online data. you can get the `P-V` and `P-CA` curves, Heat balance data and other data by clicking START. Adjust the dynamometer load control for varying the torque.
4. To get the fuel meter reading , click FC Meter Start and simultaneously open the valve of the fuel pipe.
5. When both the indicators turn light green, close the valve and you get the reading in seconds.
6. You can save data for 5 loads namely LOAD 1, LOAD 2, and so on. Finally , you can go in history and check the previously saved data.

# TURNING MOMENT DIAGRAM (CRANK EFFORT DIAGRAM)

It is defined an area under torque and angle of rotation of crank shaft which give the energy started or given by the flywheel and is used to find fluctuation of energy and hence in the design of flywheel.




**FUNCTION OF FLYWHEEL** :-

- Energy buffer (absorbing and releasing energy when required)
- Smooth motion : sudden speed drops
- Maintain momentum
- Engine balancing

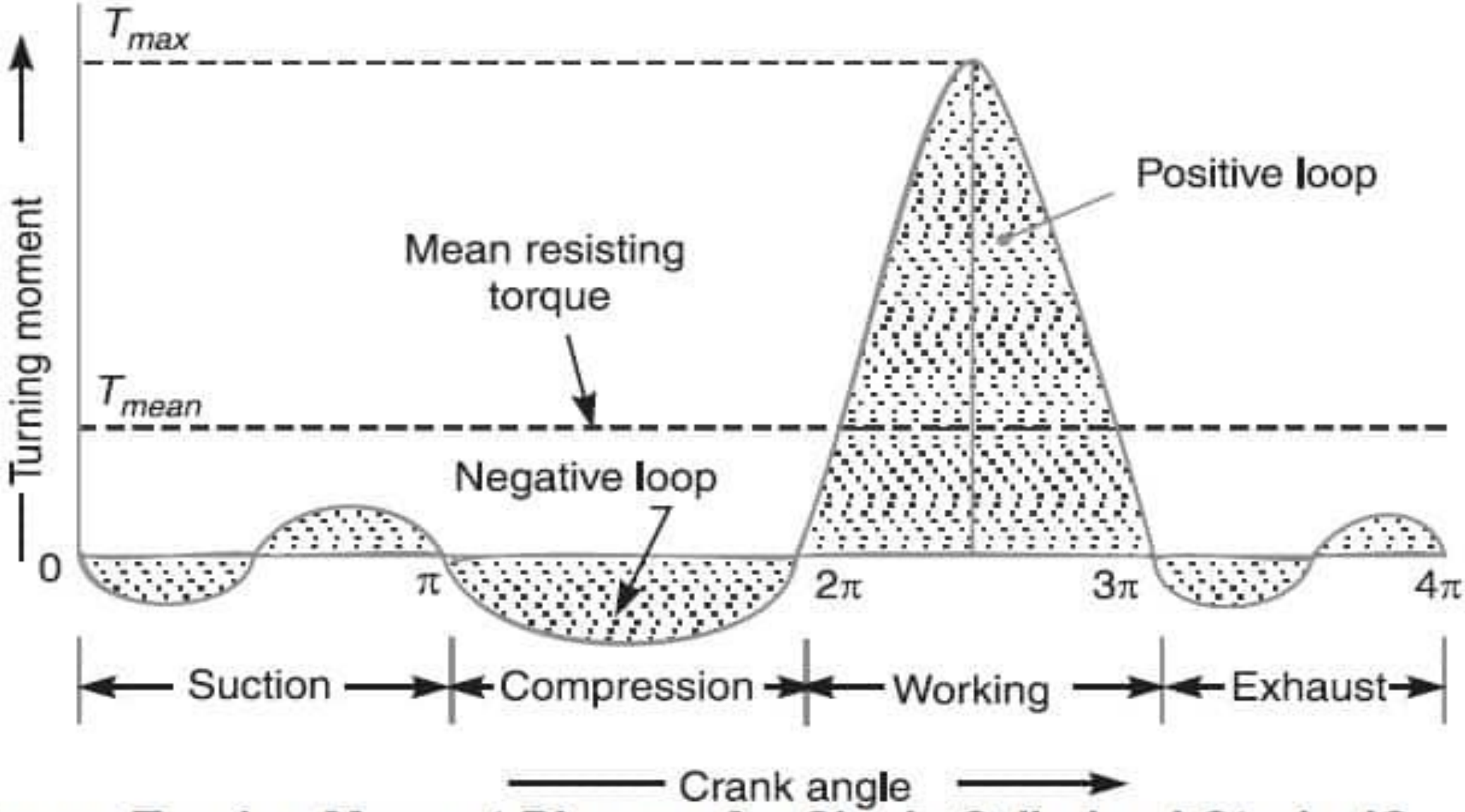
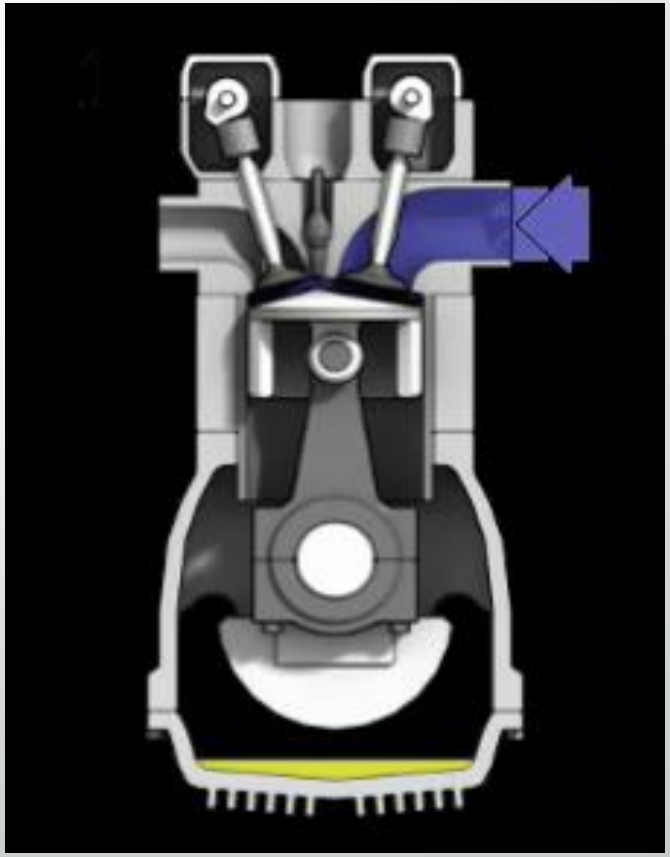



**Figure: Turning Moment Diagram for SIngle Cylinder 4 Stroke IC ENgine**

# TURNING MOMENT DIAGRAM FOR SINGLE CYLINDER 4 STROKE S.I ENGINE

[1] Since the pressure inside the engine cylinder is less than the atmospheric pressure during the suction stroke, therefore a negative loop is formed as shown in Fig.
[2] During the compression stroke, the work is done on the gases; therefore a higher negative loop is obtained.
[3] During the expansion or working stroke, the fuel burns and the gases expand; therefore a large positive loop is obtained. In this stroke, the work is done by the gases.
[4] During the exhaust stroke, the work is done on the gases; therefore a negative loop is formed.



Torque = Work done by the engine/crank angle
Work done by the cycle = Integration of `P.dV`
The total work done is angle x torque so in this case we have:

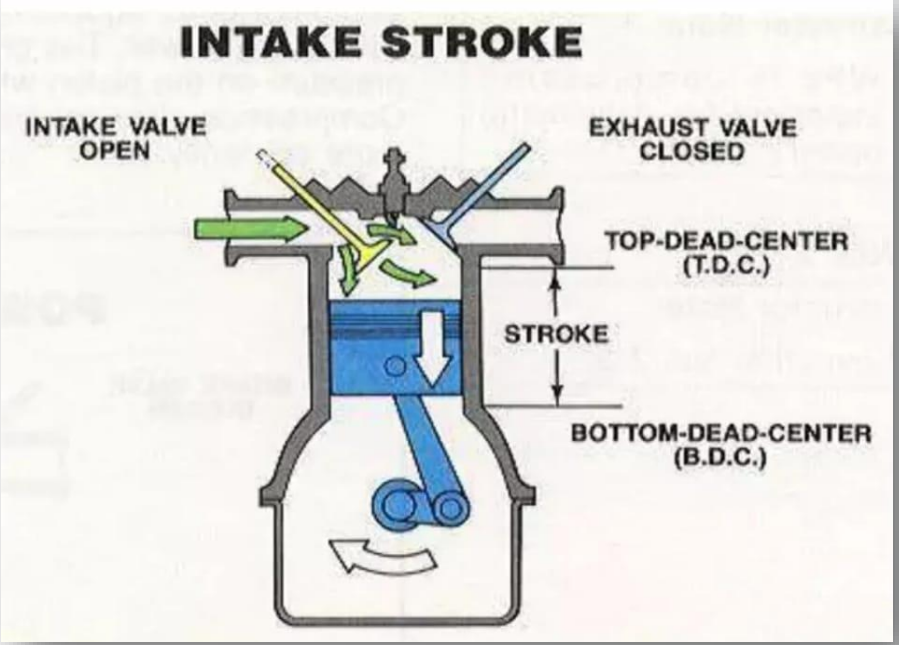
`W = T_(mean)*crank angle

crank angle = 4 π

**Note that the angle is not always `4π ` but whatever angle corresponds to the cycle. For example it would be `2π ` for a 2 stroke engine**

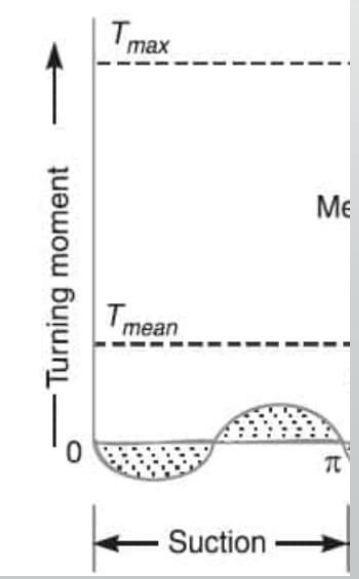
# CYCLE ANALYSIS – WHAT IS HAPPENING EXACTLY ?

## (Breakdown of the 720° cycle)

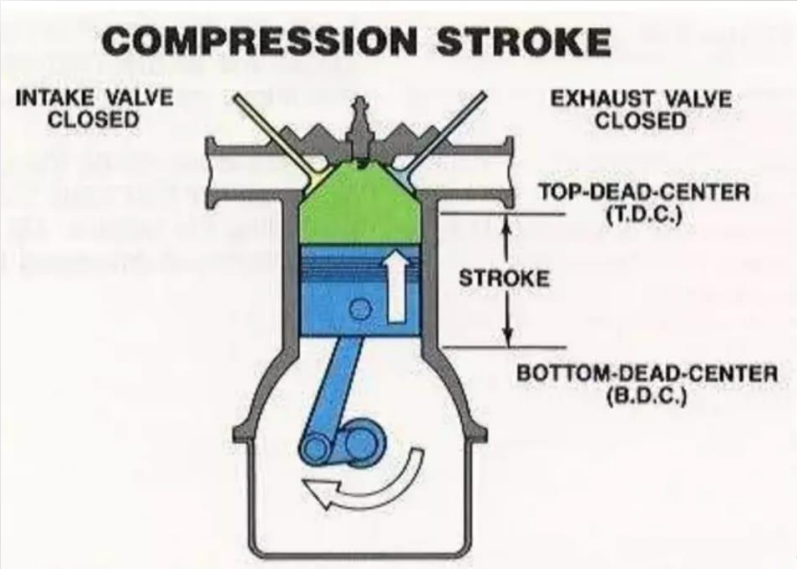



## 1. SUCTION / INTAKE STROKE (0° – 180°)

- In suction stroke piston starts at Top Dead Centre **(TDC)** of the cylinder and moves to the Bottom Dead Centre **(BDC).**
- Outlet Valve will be closed and inlet valve will be open to allowing the fresh charge of mixed fuel and air into the cylinder.
- CURVE : Slightly **Negative**
- **REASON :** Pressure inside cylinder < Atmospheric pressure. The crank has to drag the piston down (Pumping Loss).



## 2. COMPRESSION STROKE (180° – 360°)



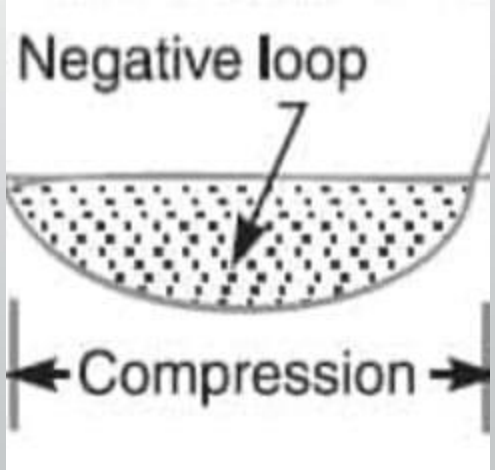

❖In compression stroke, Once piston reaches **BDC** and moves back **TDC**, inlet valve will be closed.

❖As the piston moves towards TDC.

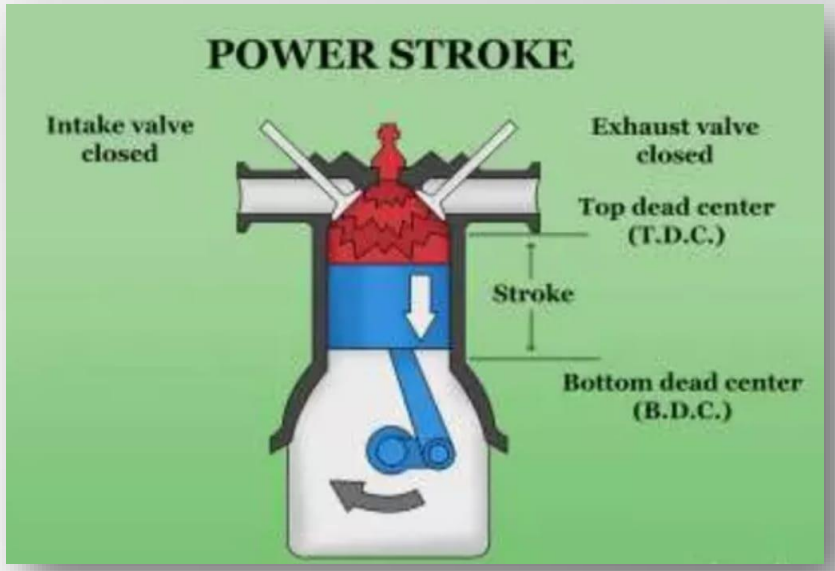
❖It compress air inside the cylinder and compression takes place. Hence it is called compression stroke.

❖CURVE : Highly **Negative**

❖**REASON :** Work is done on the gases to compress them. The flywheel supplies this energy.



# 3. EXPANSION / POWER STROKE (360° – 540°)

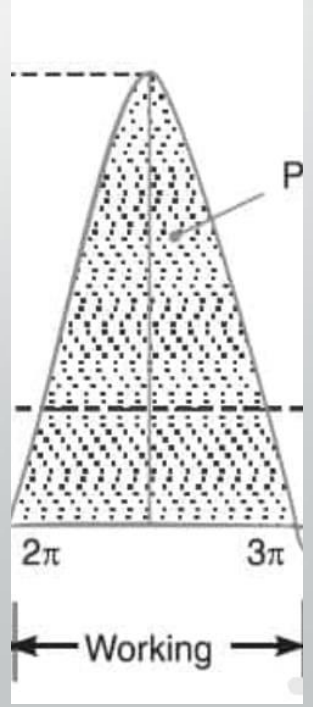



❖In expansion stroke, Both the valves are closed. When piston reaches top of its stroke, the fuel is sprinkled by the Fuel Injector and the fuel mixture is ignited due to high temperature and pressure generated inside the cylinder and push down the piston to BDC.

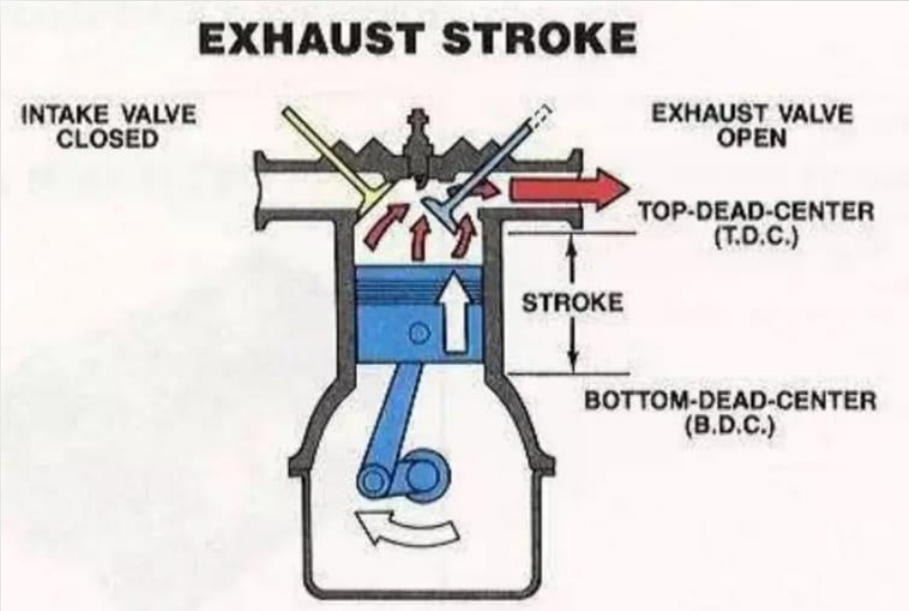
❖Hence it is known as Power generated in this stroke is stored in the flywheel for its further utilization in the other strokes.

❖ CURVE : High **Positive Peak**

❖**REASON :** This is the only stroke producing power. The area under this curve is the work done by the engine.



# 4. EXHAUST STROKE (540° – 720°)



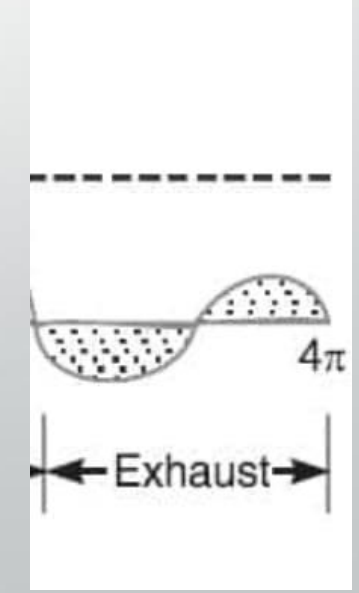

❖In this stroke exhaust valve is opened when piston reaches to BDC and moves to upward.

❖Piston pushes out the burnt gases to the atmosphere through the exhaust valve.

❖Hence called exhaust stroke and the engine is ready to begin the cycle again.

❖CURVE : Slightly **Negative**

❖**REASON :** Piston works against back-pressure to push gases out.



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# Torque vs Crank Angle Diagram

Max Torque: 100 Mean Torque: 40 Negative Torque: -30

# SIMULATION

## OBJECTIVE :

The objective of this simulation is to analyze the variation of torque with respect to the crank angle under different operating conditions. It helps in understanding the performance and efficiency of the system during a complete cycle. The results validate the accuracy and reliability of the proposed model.

SIMULATION LINK

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•**Ganesan, V.**
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•https://learnmech.com/turning-moment-diagram
•Remote Triggered Virtual Lab on Automotive Systems :
https://share.google/oZqSCGzdxnFlrjij9